PES UNIVERSITY

UE24CS320A - Capstone Project

VerdictTrace

Tamper-Evident, Quorum-Based Cross-Layer Drift Attribution for Connected Vehicle Fleets

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Batch 51

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Phase I / Review 1

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Problem Statement

Modern connected vehicles use Controller Area Network (CAN) internally and Vehicle-to-Everything (V2X) for external communication. Software faults, legitimate operational changes and malicious attacks can produce similar abnormal behaviour, complicating identification and investigation.

1. Difficulty Identifying the Actual Cause

CAN injection, falsified V2X messages and legitimate software changes can look alike.
An unusual event alone does not establish its underlying cause.

2. Unreliable and Conflicting Vehicle Information

Compromised sources may send misleading reports. Missing, delayed or conflicting CAN/V2X observations weaken security assessments and fleet corroboration.

3. Lack of Independently Verifiable Evidence

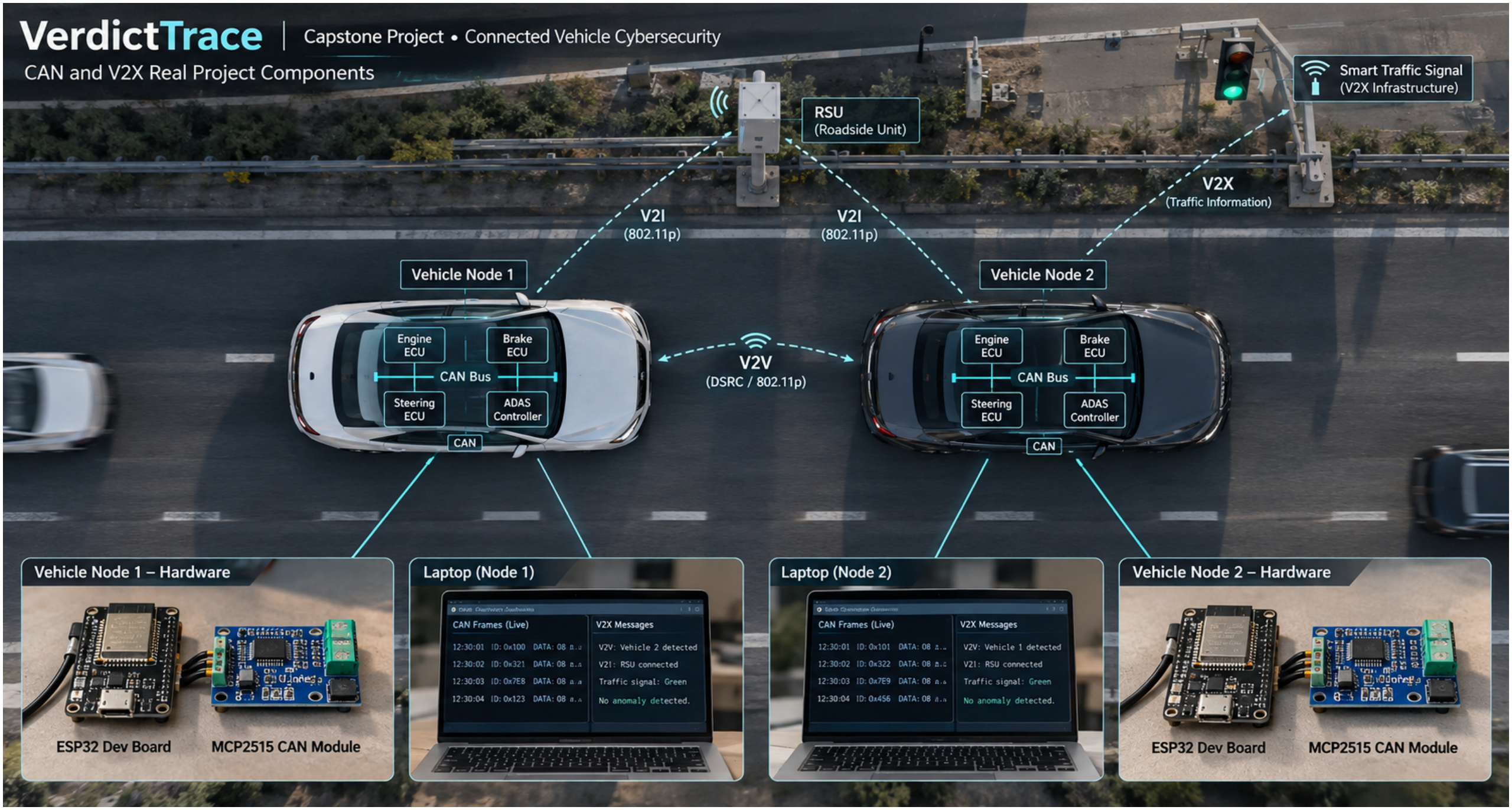
Records may be altered, deleted or unavailable during an incident or outage, making forensic reconstruction and verification difficult.

Core Research Problem

How can connected vehicles distinguish benign abnormalities from malicious attacks, identify supported causes and preserve independently verifiable evidence for investigation?

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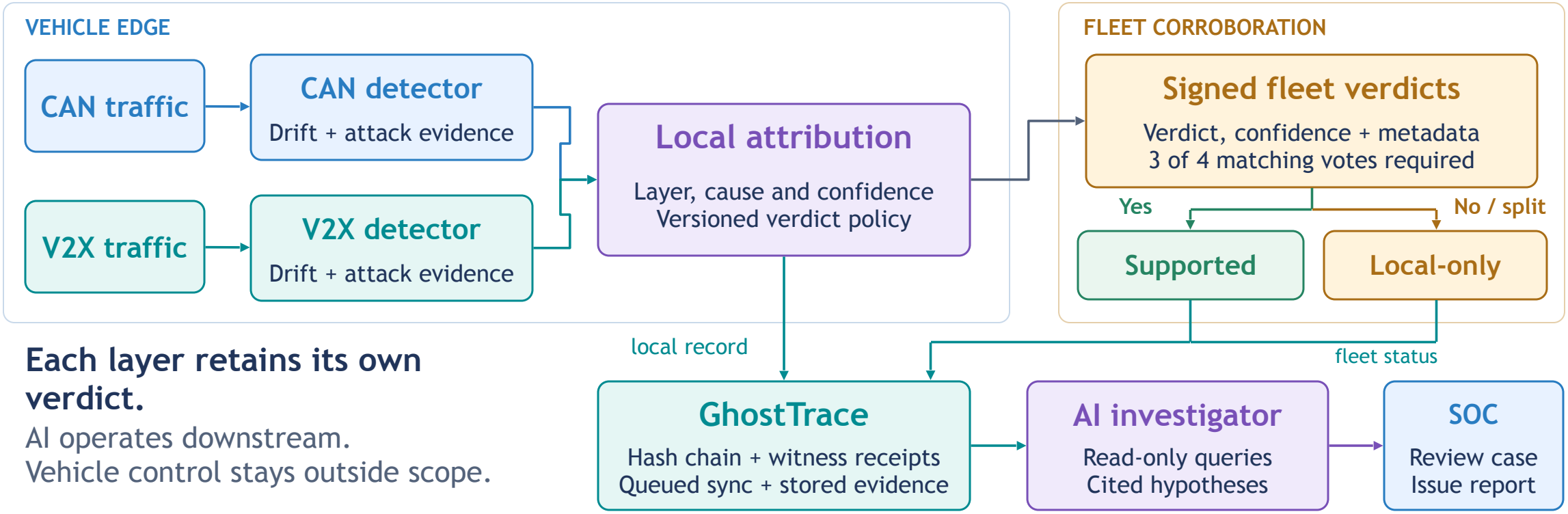
CAN and V2X Real Project Components



Concept illustration: planned CAN bench hardware and laptop-simulated V2X.

Proposed Architecture

Independent CAN and V2X detection with verified evidence for investigation



Each layer retains its own verdict.

AI operates downstream.
Vehicle control stays outside scope.

Related Work - CAN and V2X

Prior study	Established work	Our question to test
Shen et al. (2026) [1]	Separate CAN and V2X classification on synthetic data	Does matched cross-layer evidence improve cause attribution?
CND-IDS (2025) [2]	Continual novelty detection	Can benign-change false alarms fall without losing attack recall?
Golam et al. (2025) [3]	CAN/V2X/GPS agents and distributed threat response	Do independent local verdicts and source reliability add value?
ROAD (2024) [5]	Real CAN intrusion benchmark	Establish a CAN baseline. No synchronized V2X stream.
Korycki & Krawczyk (2023) [7]	Poisoning can mimic legitimate concept drift	Test matched fault/attack cases with explicit unknown outcomes.

Scope and Feasibility Study

CAN + V2X Project Scope

Independent Detection
Preserve alerts from CAN and simulated V2X.

Cause Attribution
Distinguish faults, benign drift and attacks where matched evidence permits.

Trust-Aware Fusion
Check source reliability and conflicts.
Return “insufficient evidence” when needed.

Bounded Prevention
Authorised CAN bench gateway / simulated V2X receiver filtering only. Measure false blocks.

Fleet Corroboration
Four simulated clients share signed event votes.
Local verdicts persist below quorum.

Feasibility & Resources

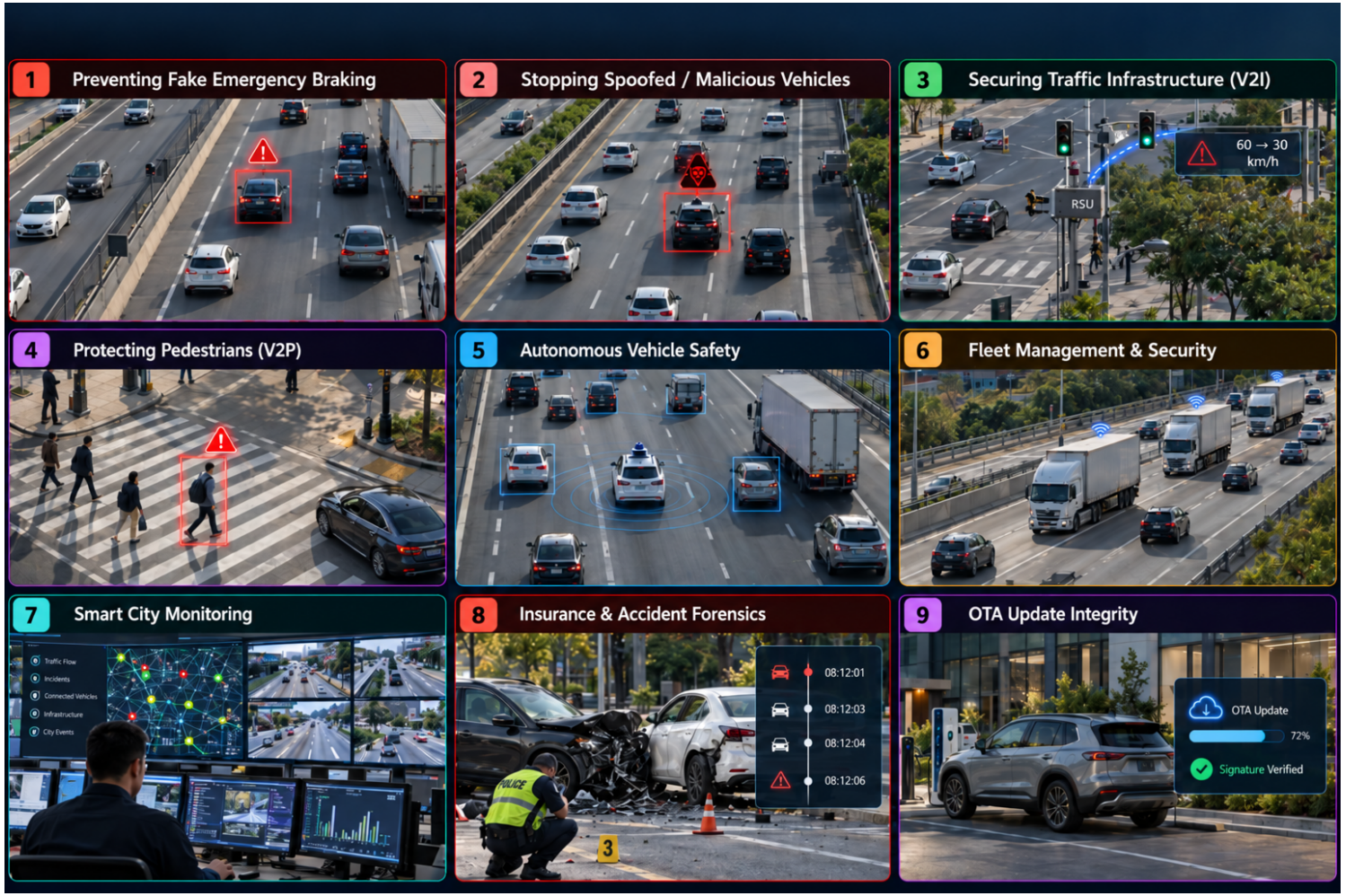
Available compute + CAN hardware
2 ESP32 + 2 MCP2515 modules, wired and terminated bus cables .

Under ₹2,000: core-board estimate
The cost mainly comes from ESP 32 boards , CAN components and API keys for better output.

Research data + test environment
ROAD (2024), CICIoV2024, VeReMi NextGen (2026) + self generated simulations.

AI API Keys
NVIDIA development access and eligible Gemini free tier. Optional paid Claude Haiku API.
Core detection and evidence checks work offline.

CAN and V2X - Applications / Use Cases



Illustrative application scenarios for proposed laboratory evaluation.

Implementation Challenges

CAN-V2X Time Alignment

CAN and V2X events may have different timestamps, making the same incident harder to match.

CAN Test-Bench Setup

ESP32 and MCP2515 integration needs reliable wiring, bus cable termination and stable message capture.

Realistic Scenario Creation

Public CAN and V2X datasets are usually independent. Linked fault-and-attack examples need controlled simulation.

Cross-Layer and Fleet Integration

Detector outputs and signed vehicle reports require careful event matching and integration testing.

Background

Why this problem

Connected vehicles exchange data over CAN and V2X, and their behavior legitimately changes with software updates and driving conditions [3].
Attackers can disguise manipulation as that normal change, so detectors confuse drift with attack [2, 7].
Analysts also need evidence they can verify, not just an alert, to decide which cause is supported [4, 9, 10].

Domain context

Networks	CAN links in-vehicle control units [5]; V2X links vehicles [3].
IoT / CPS	A connected vehicle is a sensing and control system [9].
IPS / ML	Drift in normal traffic can resemble an attack [7, 10].
Forensics	Analysts need verifiable evidence to explain events [4, 9].

Research Gap and Proposed Contribution

Cause-aware attribution [1, 2, 7, 10]

Similar anomalies can arise from faults, benign changes or attacks.

Planned evaluation: Test matched CAN/V2X events with known causes and an unknown outcome.

Trust-aware conflict handling [1, 3]

Correlated observations can still be delayed, missing or misleading.

Planned evaluation: Preserve independent alerts and test reliability-aware confidence updates.

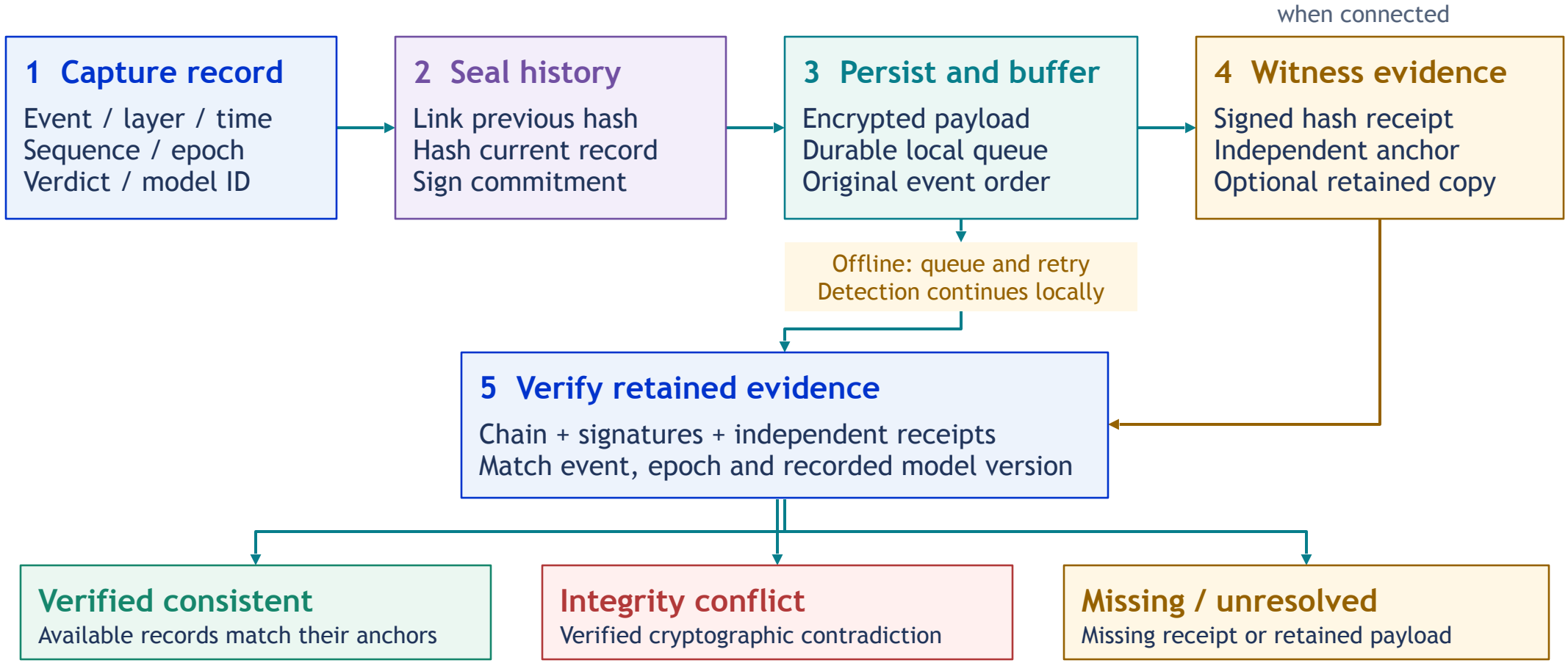
Bounded filtering and fleet robustness [3, 6]

A supported local alert may lack reliable fleet corroboration.

Planned evaluation: Measure lab false blocks and corroboration under one faulty simulated client.

GhostTrace: Evidence Preservation

Proposed evidence path: seal locally, witness independently, verify before investigation



GhostTrace: Forensics Command Centre



Concept illustration: displayed metrics are illustrative, not experimental results.

GhostTrace: Motivation, Prior Work & Research Gap

Motivation: incomplete incident histories weaken investigation

NHTSA's 2024 Cruise action concerned incomplete reports of a 2023 pedestrian crash.
It illustrates reporting and reconstruction needs, without establishing raw-log tampering.

Prior research	Contribution	Question for GhostTrace
Nitro (2025) [4]	Tamper-evident system auditing	How does verification behave during vehicle connectivity loss?
Accountability of Things (2023, preprint) [8]	Witnessed logging for devices	Can independent receipts expose conflicts without labelling every gap as tampering?
Sharma & Gillanders (2022) [9]	CAV cybersecurity and forensic review	Can retained incident histories support repeatable, verifiable reconstruction?

Research gap to test

Distinguish verified alteration from delayed or unavailable evidence, and measure what independent copies can recover. Support collision and insurance incident review.

GhostTrace: Scope, Feasibility & Challenges

Evidence-preservation scope

Hash-linked event history and signed checkpoints.
Independent receipts and delayed synchronisation.
Optional encrypted copies for later retrieval.
Verification states with explicit unresolved gaps.

Feasibility

Use existing laptops, CAN/V2X scenario records and four simulated clients.
Separate evidence verification from AI narration. No extra vehicle hardware.

Engineering challenges

Offline witness delay

Receipts may arrive after the event.

Witness identity and replay

Bind signatures to signer, event and epoch.

Missing vs changed records

Absence alone cannot establish alteration.

Storage and recovery

Hashes cannot restore deleted payloads.

GhostTrace: Forensic Evaluation

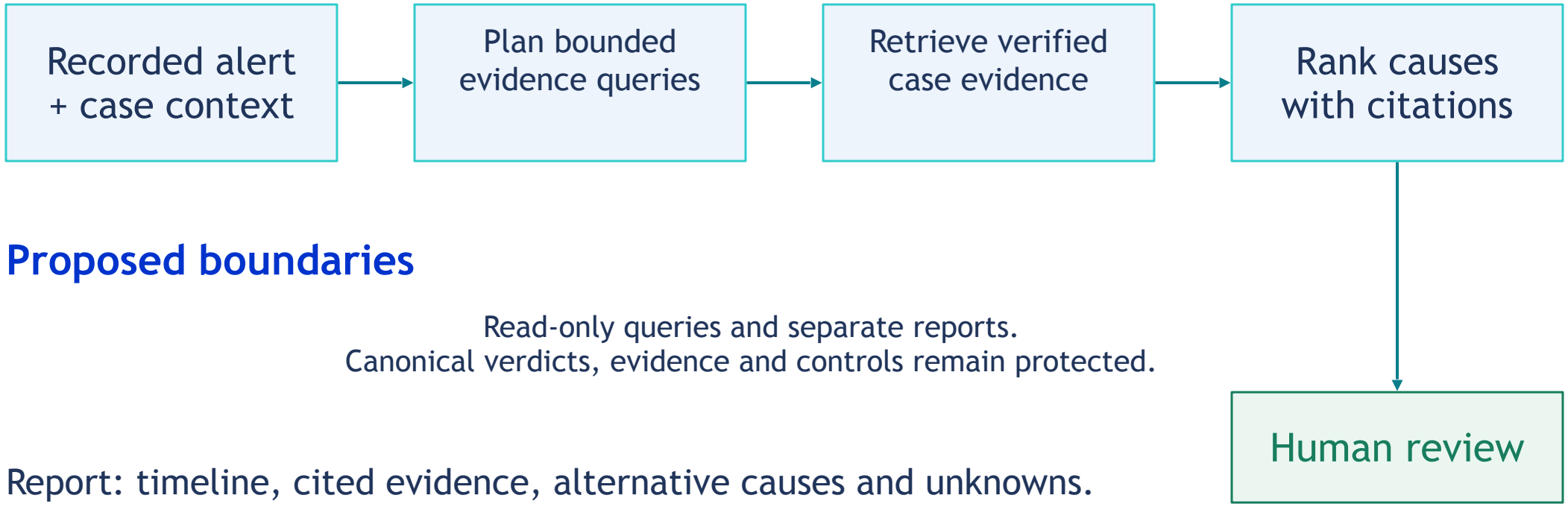
Experiment	Question / planned measure
Altered record or hash conflict	Verified integrity-conflict detection and tamper recall
Deletion, rollback or replay	Detection of missing history and continuity violations
Connectivity outage or delayed receipt	False tamper allegations and unresolved-gap reporting
Forged or mismatched witness receipt	Signature, identity, event and epoch validation
Lost payload with an independent copy	Recoverable-event fraction and retrieval latency

Baselines

Local records / hash chain / chain + receipts / receipts + independent copies

Bounded Agentic AI Investigation

The agent selects evidence queries and produces a cited case report



Proposed boundaries

Read-only queries and separate reports.
Canonical verdicts, evidence and controls remain protected.

Report: timeline, cited evidence, alternative causes and unknowns.

Dataset Construction and Evaluation

Public benchmarks

ROAD (2024) [5]

Real CAN and labelled attacks, plus simulated masquerade variants.

CICIoV2024 (2024)

CAN spoofing / DoS from a real, immobilised vehicle.

VeReMi NextGen (2026)

Simulated urban / highway V2X. Contains no paired CAN stream.

Data we will generate

Synchronized CAN + V2X scenarios

Shared event IDs, clocks and known causes. Benign faults/drift, attacks and delayed inputs.

ESP32 + MCP2515 CAN recordings

Physical bench frames and capture timing. Controlled bus, without real-vehicle actuation.

Dedicated GhostTrace experiments: Slide 16

Planned comparison	Metrics
Independent detection vs timing / trust-aware fusion	Attack recall, false alarms, cause macro-F1
Aligned vs delayed / missing / conflicting inputs	Attribution, abstention, latency and memory
Detection-only vs authorised lab filtering	False blocks and filtering latency
Local vs four-client fleet corroboration	False corroboration and below-quorum behaviour

Expected Deliverables

Phase I

Research and baseline

01 Threat model and scenario labels documented.

02 CAN / V2X baselines run on ROAD [5] / scenario data.

03 False alarms and attack recall reported.

Phase II

Integrated prototype

01 Timing compared on cause accuracy.

02 Fleet + GhostTrace tested on fault / outage cases.

03 AI + SOC demonstrate cited, read-only investigation.

Phase III

Evaluation and release

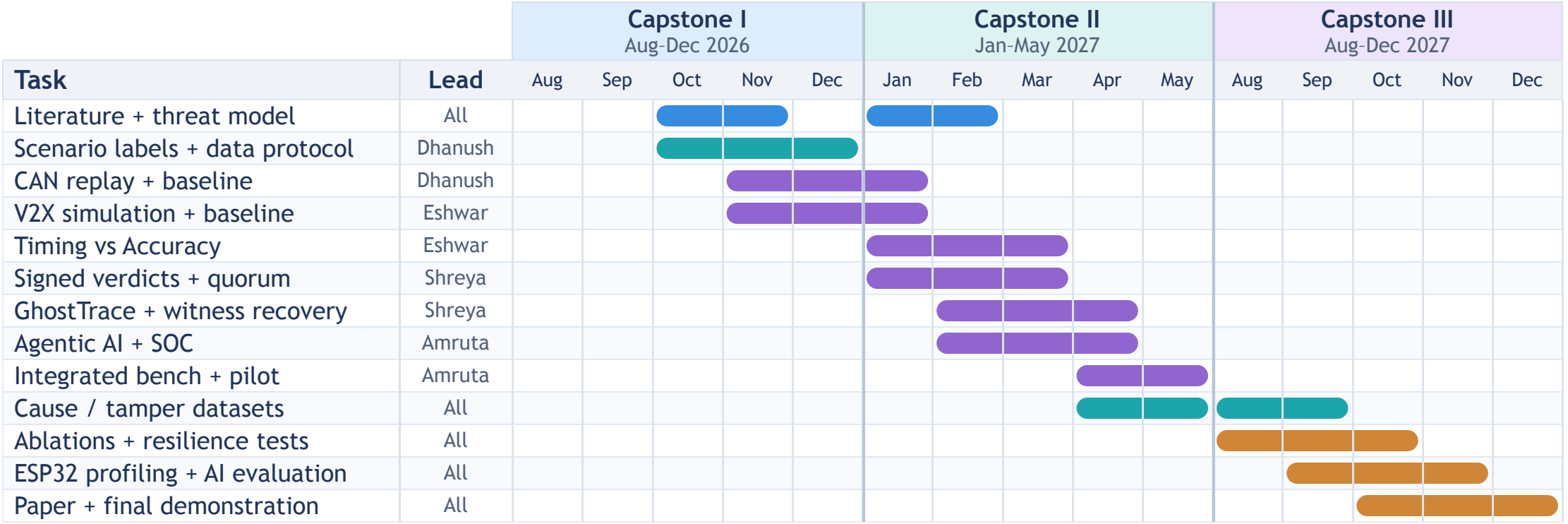
01 Both datasets released with provenance, labels and splits.

02 Ablations report recall, cause accuracy and ESP32 workload.

03 Re-runnable code, report and paper prepared.

Project Timeline and Team Responsibilities

Proposed plan from research to evaluated prototype



References

References ordered by publication year, newest first; paper titles link to sources

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Securing autonomous vehicles: a dual-domain intrusion detection system for intra-vehicle and external networks

Peer-to-Peer Networking and Applications, 2026

[2] S. Fuhrman, O. Gungor and T. Rosing

CND-IDS: Continual Novelty Detection for Intrusion Detection Systems

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[3] M. Golam, J.-M. Lee and D.-S. Kim

Language Agent Model-Driven Distributed Consensus for Advanced IoV Threat Response Management

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[4] R. Zhao et al.

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ACM CCS, 2025

[5] M. E. Verma et al.

A comprehensive guide to CAN IDS data and introduction of the ROAD dataset

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[6] X. Sui, S. Duan and H. Zhang

BG: A Modular Treatment of BFT Consensus Toward a Unified Theory of BFT Replication

IEEE TIFS, vol. 19, pp. 44–58, 2024

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IEEE Access, 2022

[10] L. Yang et al.

CADE: Detecting and Explaining Concept Drift Samples for Security Applications

USENIX Security, 2021

[11] P. K.C., R. Soman and P. Honnavalli

Validity of Forensic Evidence using Hash Function

IEEE ICCES, 2020

Thank You

Questions and Discussion

VerdictTrace

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